In [2]:

```python
import numpy as np
#for numerical computing for arrya operation and mathematical function liner algebra
import pandas as pd
#using for data manipulation and analysis easy to use data structure and analysis
from sklearn.model_selection import train_test_split
#Provides a wide range of tools and algorithms for various ML task includeing classification regression and clustering
#train test slipt function help in splitting the data set into training and the testing subset
from sklearn.feature_extraction.text import TfidfVectorizer
#TFID Vector feature of extraction commonly usedd in NLP and text mining task convert of text doc into a numerical representation
from sklearn.linear_model import LogisticRegression
# Classification algorithm used in ML suitable for binary classification problem
from sklearn.metrics import accuracy_score
#accuracy function is a performance used for evalute the performance for classification by measuring the accuracy of the prediction
```

In [19]:

```python
df = pd.read_csv(r'C:\Users\sahil\OneDrive\Documents\mail_data.csv')
```

In [20]:

```python
df.head()
```

Out[20]:

	Category	Message
0	ham	Go until jurong point, crazy.. Available only ...
1	ham	Ok lar... Joking wif u oni...
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...
3	ham	U dun say so early hor... U c already then say...
4	ham	Nah I don't think he goes to usf, he lives aro...

In [21]:

```python
print(df)
```

```
     Category                                            Message
0         ham  Go until jurong point, crazy.. Available only ...
1         ham                      Ok lar... Joking wif u oni...
2        spam  Free entry in 2 a wkly comp to win FA Cup fina...
3         ham  U dun say so early hor... U c already then say...
4         ham  Nah I don't think he goes to usf, he lives aro...
...       ...                                                ...
5567     spam  This is the 2nd time we have tried 2 contact u...
5568      ham               Will ü b going to esplanade fr home?
5569      ham  Pity, * was in mood for that. So...any other s...
5570      ham  The guy did some bitching but I acted like i'd...
5571      ham                         Rofl. Its true to its name

[5572 rows x 2 columns]
```

In [23]:

```python
data = df.where((pd.notnull(df)), '')
#converting this mail data into normal
```

In [24]:

```python
data.head()
#printing first few line
```

Out[24]:

	Category	Message
0	ham	Go until jurong point, crazy.. Available only ...
1	ham	Ok lar... Joking wif u oni...
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...
3	ham	U dun say so early hor... U c already then say...
4	ham	Nah I don't think he goes to usf, he lives aro...

In [25]:

```python
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5572 entries, 0 to 5571
Data columns (total 2 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Category  5572 non-null   object
 1   Message   5572 non-null   object
dtypes: object(2)
memory usage: 87.2+ KB
```

In [27]:

```python
data.loc[data['Category'] =='spam','Category',]=0
data.loc[data['Category'] =='ham','Category',]=1
#if 0 will show as the spam and 1 will show as the ham mail
```

In [28]:

```python
x = data['Message']

y = data['Category']
```

In [29]:

```python
print(x)
```

```
0       Go until jurong point, crazy.. Available only ...
1                           Ok lar... Joking wif u oni...
2       Free entry in 2 a wkly comp to win FA Cup fina...
3       U dun say so early hor... U c already then say...
4       Nah I don't think he goes to usf, he lives aro...
                              ...
5567    This is the 2nd time we have tried 2 contact u...
5568                 Will ü b going to esplanade fr home?
5569    Pity, * was in mood for that. So...any other s...
5570    The guy did some bitching but I acted like i'd...
5571                           Rofl. Its true to its name
Name: Message, Length: 5572, dtype: object
```

In [30]:

```python
print(y)
```

```
0       1
1       1
2       0
3       1
4       1
       ..
5567    0
5568    1
5569    1
5570    1
5571    1
Name: Category, Length: 5572, dtype: object
```

In [33]:

```python
x_train, x_test, y_train, y_test = train_test_split(x,y, test_size=0.2, random_state = 3)
#0.2 means it will be 80 and 20% and 80% will be training and 20% will be test/
#and random state is the hyperparameter used to control any randomness invlove in ML model
```

In [34]:

```python
print(x.shape)
print(x_train.shape)
print(x_test.shape)
```

```
(5572,)
(4457,)
(1115,)
```

In [35]:

```python
print(y.shape)
print(y_train.shape)
print(y_test.shape)
```

```
(5572,)
(4457,)
(1115,)
```

In [42]:

```python
feature_extraction = TfidfVectorizer(min_df = 1, stop_words = 'english', lowercase=True)

x_train_features = feature_extraction.fit_transform(x_train)
x_test_features = feature_extraction.transform(x_test)

y_train = y_train.astype('int')
y_test = y_test.astype('int')

#transforming the test data to the feature vector
#stop words are the english words that dose not add much meaning to the sentence they can easy ignored
#wihtout sacrificing the meaning of the sentence for eg: words like the, he, has, have
#
```

In [43]:

```python
print(x_train)
```

```
3075                  Don know. I did't msg him recently.
1787    Do you know why god created gap between your f...
1614                         Thnx dude. u guys out 2nite?
4304                                      Yup i'm free...
3266    44 7732584351, Do you want a New Nokia 3510i c...
                              ...
789     5 Free Top Polyphonic Tones call 087018728737,...
968     What do u want when i come back?.a beautiful n...
1667    Guess who spent all last night phasing in and ...
3321    Eh sorry leh... I din c ur msg. Not sad alread...
1688    Free Top ringtone -sub to weekly ringtone-get ...
Name: Message, Length: 4457, dtype: object
```

In [44]:

```python
print(x_train_features)
```

```
<Compressed Sparse Row sparse matrix of dtype 'float64'
	with 34775 stored elements and shape (4457, 7431)>
  Coords	Values
  (0, 2329)	0.38783870336935383
  (0, 3811)	0.34780165336891333
  (0, 2224)	0.413103377943378
  (0, 4456)	0.4168658090846482
  (0, 5413)	0.6198254967574347
  (1, 3811)	0.17419952275504033
  (1, 3046)	0.2503712792613518
  (1, 1991)	0.33036995955537024
  (1, 2956)	0.33036995955537024
  (1, 2758)	0.3226407885943799
  (1, 1839)	0.2784903590561455
  (1, 918)	0.22871581159877646
  (1, 2746)	0.3398297002864083
  (1, 2957)	0.3398297002864083
  (1, 3325)	0.31610586766078863
  (1, 3185)	0.29694482957694585
  (1, 4080)	0.18880584110891163
  (2, 6601)	0.6056811524587518
  (2, 2404)	0.45287711070606745
  (2, 3156)	0.4107239318312698
  (2, 407)	0.509272536051008
  (3, 7414)	0.8100020912469564
  (3, 2870)	0.5864269879324768
  (4, 2870)	0.41872147309323743
  (4, 487)	0.2899118421746198
  :	:
  (4454, 2855)	0.47210665083641806
  (4454, 2246)	0.47210665083641806
  (4455, 4456)	0.24920025316220423
  (4455, 3922)	0.31287563163368587
  (4455, 6916)	0.19636985317119715
  (4455, 4715)	0.30714144758811196
  (4455, 3872)	0.3108911491788658
  (4455, 7113)	0.30536590342067704
  (4455, 6091)	0.23103841516927642
  (4455, 6810)	0.29731757715898277
  (4455, 5646)	0.33545678464631296
  (4455, 2469)	0.35441545511837946
  (4455, 2247)	0.37052851863170466
  (4456, 2870)	0.31523196273113385
  (4456, 5778)	0.16243064490100795
  (4456, 334)	0.2220077711654938
  (4456, 6307)	0.2752760476857975
  (4456, 6249)	0.17573831794959716
  (4456, 7150)	0.3677554681447669
  (4456, 7154)	0.24083218452280053
  (4456, 6028)	0.21034888000987115
  (4456, 5569)	0.4619395404299172
  (4456, 6311)	0.30133182431707617
  (4456, 647)	0.30133182431707617
  (4456, 141)	0.292943737785358
```

In [45]:

```python
model = LogisticRegression()
```

In [47]:

```python
#training data
model.fit(x_train_features, y_train)
```

Out[47]:

▾ LogisticRegression

▶ Parameters

In [57]:

```python
#the evaluting of the training model and the prediction of the tarin dfata
prediction_on_training_data = model.predict(x_train_features)
accuracy_on_training_data = accuracy_score(y_train, prediction_on_training_data)
```

In [58]:

```python
print('Acc on training data : ' , accuracy_on_training_data)
```

```
Acc on training data :  0.9676912721561588
```

In [59]:

```python
prediction_on_training_data = model.predict(x_test_features)
accuracy_on_training_data = accuracy_score(y_test, prediction_on_training_data)
```

In [60]:

```python
print('Acc on training data : ' ,  accuracy_on_training_data)
```

```
Acc on training data :  0.9668161434977578
```

In [72]:

```python
#building predicting system
input_your_mail = ["We detected unusual activity on your account. Your account will be suspended within 24 hours unless you verify your information immediately."]

input_data_features = feature_extraction.transform(input_your_mail)

prediction = model.predict(input_data_features)

print(prediction)
#if 1 this is ham mail
#if 0 this is spam mail

if prediction[0] == 1:
    print('Ham mail')

else:
    print('Spam mail')
```

```
[1]
Ham mail
```

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]: